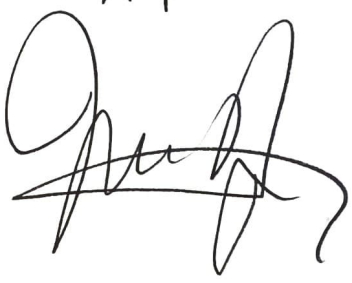
On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and to subject to harsh discipline issues.

150719 691

Merve Sena Arpacik



Q.1. Consider the matrix $A = \begin{bmatrix} 1 & \alpha & -4 & \alpha \\ 3\alpha & 5 & -9 & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$

a) Find the unknown entries α, γ, β, k and m=?

b) Find the eliminated upper triangle matrix U = ?

c) Determine the remaining elimination matrice.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}$$

2) A geometry in 'k' dimensional space is given by $x = 5y$ expression, where 'x' and 'y' represents the axis. The intersection with another geometry gives a volume.

a) Give the minimum dimension of the space with your valuable comments

b) Propose an expression for the unknown geometry to have a volume in intersection.

c) If proposed a geometry $x + 4y - z = 6$. Find all vectors in the intersection. Do these vectors form a vector space?. The intersection has how many dimensional space?

3) Suppose u is a unit vector in R^n. This Problem is about the n by n symmetric matrix $H = I - 2uu^T$:

a) Show directly that $H^2 = I$

b) Show that H is an orthonormal matrix.

c) One eigenvector of H is u itself. Find the corresponding eigenvalue.

d) Is H diagnolizable? why or why not?

a) $H^2 = H \cdot H = (I - 2uu^T)(I - 2uu^T)$

$= I^2 - I \cdot 2uu^T - 2uu^T \cdot I + 4uu^T \cdot uu^T$

$= I(I - 2uu^T - 2uu^T) + 4u \cdot \|u\|^2 \cdot u^T$

$= I - 2uu^T - 2uu^T + 4u \cdot \|u\|^2 \cdot u^T$

$\|u\|^2 = u^T \cdot u$

$\|u\| = \pm\sqrt{u^T \cdot u}$

$\rightarrow \|u\| = 1$

$H^2 = I - 4uu^T + 4uu^T$

$H^2 = I - 4(1) + 4(1)$

$\boxed{H^2 = I}$

b) $HH^T = (I - 2uu^T)(I - 2uu^T)^T$

$= (I - 2uu^T)(I^T - (2uu^T)^T)$

$= (I - 2uu^T)(I - 2(u^T)^T \cdot u^T)$

$= (I - 2uu^T)(I - 2uu^T)$

$= I^2 - 2uu^T - 2uu^T + 4uu^T \cdot uu^T$

$= I - 4uu^T + 4u\|u\|^2 u^T$

$= I - 4uu^T + 4uu^T$

$= I - 4(1) + 4(1) = I$

$H^T \cdot H = I$

$H \cdot H^T = H^T \cdot H = I$

so H is an orthogonal matrix.

$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

c) $Hu = (I - 2uu^T)u$

$= Iu - 2uu^T u$

$= u - 2u\|u\|^2$

$\|u\| = 1 = \pm\sqrt{u^T.u}$

$u^T.u = \|u\|^2 = 1$

Hence, $Hu = u.2u.1 = -u$

$Hu = -u = (-1).u$

$Hu = \lambda u$ where $\lambda = -1$

So, u is an eigen vector of H, corresponding eigenvalue $uu = \lambda = -1$

d) $H = (I - 2uu^T)$

$H^T = (I - 2uu^T)^T$

$= I^T - 2(uu^T)^T$

$= I - 2(u^T)^T.u^T$

$= I - 2uu^T = H$

H is symmetric and diagonalizable matrix.

There is a theorem which states that

$H = Q \Lambda Q^T$

$= Q \Lambda . Q^{-1}$

$= (Q^T)^{-1} \Lambda Q^{-1}$

$= P^{-1} \Lambda P$

Two tanks are connected, first tank has $v(0)$ liters of liquied and the second tank has $w(0) = 40$ liters of water. The liquid diffusion between the tanks are proportional to the difference $v - w$.

a) Which one of the following differential equation set expresses the process? (v_k : the liquid diffused to the second tank, w_k : the water diffused to the second tank)

$$\begin{bmatrix} v_{k+1} \\ w_{k+1} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_k \\ w_k \end{bmatrix} \qquad \frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix}$$

$$\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix}$$

b) Determine percentages of the mixture in tank 1 at time $t = 1$ sec.

c) Find how much liters of liquids are exist in each tank in steady state?.